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**PROJECT 2 - UNSUPERVISED LEARNING
(CLUSTERING)**

DATA MINING - SECP2753-02

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1.0 Introduction

Heart disease is one of the most common and serious health issues in today's world. This disease still remains a significant threat and a burden towards the healthcare system and also affecting several individuals each year even today. The early detection to prevent heart related conditions can be crucial to improving patients outcome and reducing healthcare costs. Factors such as smoking, gender, age, blood pressure, cholesterol levels, and a family history of heart disease play a significant role in its development.

Given the nature of heart diseases as it is, traditional diagnostic methods do not always capture the complexity of individual risk profiles. Meanwhile, machine learning can provide a promising approach by using the characteristics of each dataset to uncover the hidden patterns, analyze interactions between variables to make accurate predictions.

In this project, we aim to use machine learning techniques to reveal the cluster model for each business insights. The unsupervised learning algorithms that we apply on this project include K-Means and AGNES (Agglomerative Nesting). The goal is to group abstract entities such as heart disease status into clusters of similar objects, enabling a better understanding of underlying patterns without relying on predefined labels. Before performing clustering, it is important to carry out data cleaning and preparation to ensure the dataset is accurate, consistent, and suitable for cluster analysis.

2.0 Business insight

There are several business insights that consider the best to use in the clustering. There are only a few that are chosen as the most relatable for this project.

2.1 Heart Disease Status Segmentation

This insight will show the status of getting a heart disease using all the features to identify natural clusters within the dataset. Both healthcare providers and normal citizens can benefit from the insight provided by these clusters, which can then guide preventive measures, customized insurance plans, and focused health interventions.

2.2 Lifestyle-Driven Heart Disease Risk & Preventative Program Design

This insight uncovers distinct population segments within the dataset based on the interplay of lifestyle factors such as alcohol consumption, sleep habits, and other health indicators that contribute to heart disease risk. By understanding these specific lifestyle-driven risk profiles, public health initiatives and wellness programs can design highly targeted preventative interventions, educational campaigns, and support systems to mitigate the impact of modifiable behaviors on cardiovascular health.

3.0 Data Preprocessing and Preparation

Data Preprocessing and Preparation are essential steps in preparing data for machine learning models. Data Preprocessing and preparation involves cleaning and transforming raw data into a format that can be effectively used by both clustering algorithms. This will allow the data cluster to be more accurate and meaningful.

3.1 Handling Missing Data

To prepare the dataset for classification, several preprocessing steps were performed. Missing values were handled by filling in numerical attributes with their mean and categorical attributes with their mode. For instance, attributes such as Cholesterol Level and BMI were filled with the mean, while Gender and Exercise Habits were filled with the mode.

```
# Impute missing values for numerical columns using the mean.
for col in df_processed.select_dtypes(include=['float64']).columns:
    df_processed[col] = df_processed[col].fillna(df_processed[col].mean())

# Impute missing values for categorical columns using the mode.
for col in df_processed.select_dtypes(include=['object']).columns:
    df_processed[col] = df_processed[col].fillna(df_processed[col].mode()[0])
```

Figure 3.1 Fill in the missing data for both numerical and categorical data

In the data preprocessing process, mapping categorical values to numerical representations is also essential for enabling machine learning algorithms to process the data effectively. In this case, several binary and ordinal categorical columns such as 'Gender', 'Smoking', 'Diabetes', and 'Exercise Habits' were converted into numeric format using a predefined mapping dictionary. This transformation standardizes all the categorical data to ensure that numerical models like clustering can interpret the values correctly.

```
# Map binary and ordinal categorical columns to numerical values.
mapping = {
    'Gender': {'Male': 1, 'Female': 0},
    'Smoking': {'Yes': 1, 'No': 0},
    'Family Heart Disease': {'Yes': 1, 'No': 0},
    'Diabetes': {'Yes': 1, 'No': 0},
    'High Blood Pressure': {'Yes': 1, 'No': 0},
    'Low HDL Cholesterol': {'Yes': 1, 'No': 0},
    'High LDL Cholesterol': {'Yes': 1, 'No': 0},
    'Exercise Habits': {'Low': 0, 'Medium': 1, 'High': 2},
    'Alcohol Consumption': {'None': 0, 'Low': 1, 'Medium': 2, 'High': 3},
    'Stress Level': {'Low': 0, 'Medium': 1, 'High': 2},
    'Sugar Consumption': {'Low': 0, 'Medium': 1, 'High': 2},
    'Heart Disease Status': {'Yes': 1, 'No': 0}
}
```

Figure 3.2 Predefined mapping the categorical data

3.2 Normalization

As part of the preprocessing pipeline, normalization was performed on the dataset using the StandardScaler from the sklearn.preprocessing module. Normalization is a crucial step, particularly for clustering algorithms that are sensitive to the scale of input features, such as K-Means and hierarchical clustering. This will help to prevent from being biased toward features with large numerical ranges.

```
# Normalize the data.
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df_processed)
```

Figure 3.3 Normalizing the dataset

3.3 Dimensionality Reduction

Dimensionality reduction was applied using Principal Component Analysis to reduce the complexity of the dataset. This PCA technique converts the standardized data into a lower-dimensional space that consists of only two principal components in this project. This will help to improve the computational efficiency and also help in visualizing the clustering structure in a two dimension plot. Then, the transformed data will be added to the dataframe and will serve as the final preprocessed input for clustering analysis.

```
# Apply PCA for dimensionality reduction to 2 components.
pca = PCA(n_components=2)
pca_data = pca.fit_transform(scaled_data)

# Create a new dataframe with PCA components.
df_pca = pd.DataFrame(data=pca_data, columns=['PC1', 'PC2'])
```

Figure 3.4 Reduce Dimensionality using PCA and finalize preprocessed data

4.0 Clustering Algorithm

For a numerous types of algorithm for clustering, we use K-means clustering and Agglomerative Nesting (AGNES) as our choice of algorithm in this project. Each of these models will be used in the Python programming language throughout this project.

4.1 K-means Clustering

K-means clustering is an unsupervised learning algorithm that groups unlabeled data into distinct clusters based on similarity. It requires the user to specify the number of clusters, k , in advance, and then assigns data points to the nearest cluster centroid, updating the centroids iteratively. This simple and efficient method is widely used in tasks like market segmentation and image compression [3]. For this project, we used the k value of 3 which is determined using the elbow method to find the optimal k value. Below are the figures of code usage for the elbow method implemented in python and the graph.

```
sse = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(df_pca[['PC1', 'PC2']])
    sse.append(kmeans.inertia_)

# Plot the Elbow graph
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), sse, marker='o')
plt.title('Elbow Method for K-Means')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('SSE')
plt.grid(True)
plt.xticks(np.arange(1, 11, 1))
plt.tight_layout()
```

Figure 4.1 Python code Elbow Method

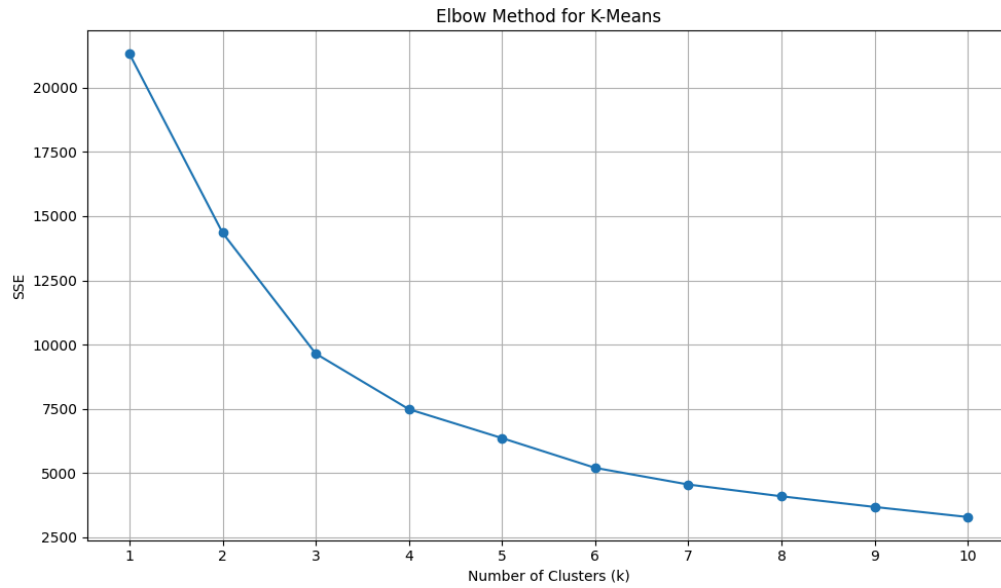


Figure 4.2 Elbow Method graph

After this method, we applied the k value to the k-means clustering model. Figure below is the example code of implementing the k-mean in python and the result of the clustering.

```
pca_for_kmeans = df_pca[['PC1', 'PC2']].copy()
kmeans = KMeans(n_clusters=k_optimal, random_state=42, n_init=10)
kmeans_labels = kmeans.fit_predict(pca_for_kmeans)
df_pca['kmeans_cluster'] = kmeans_labels
```

Figure 4.3 Python code for k-means clustering

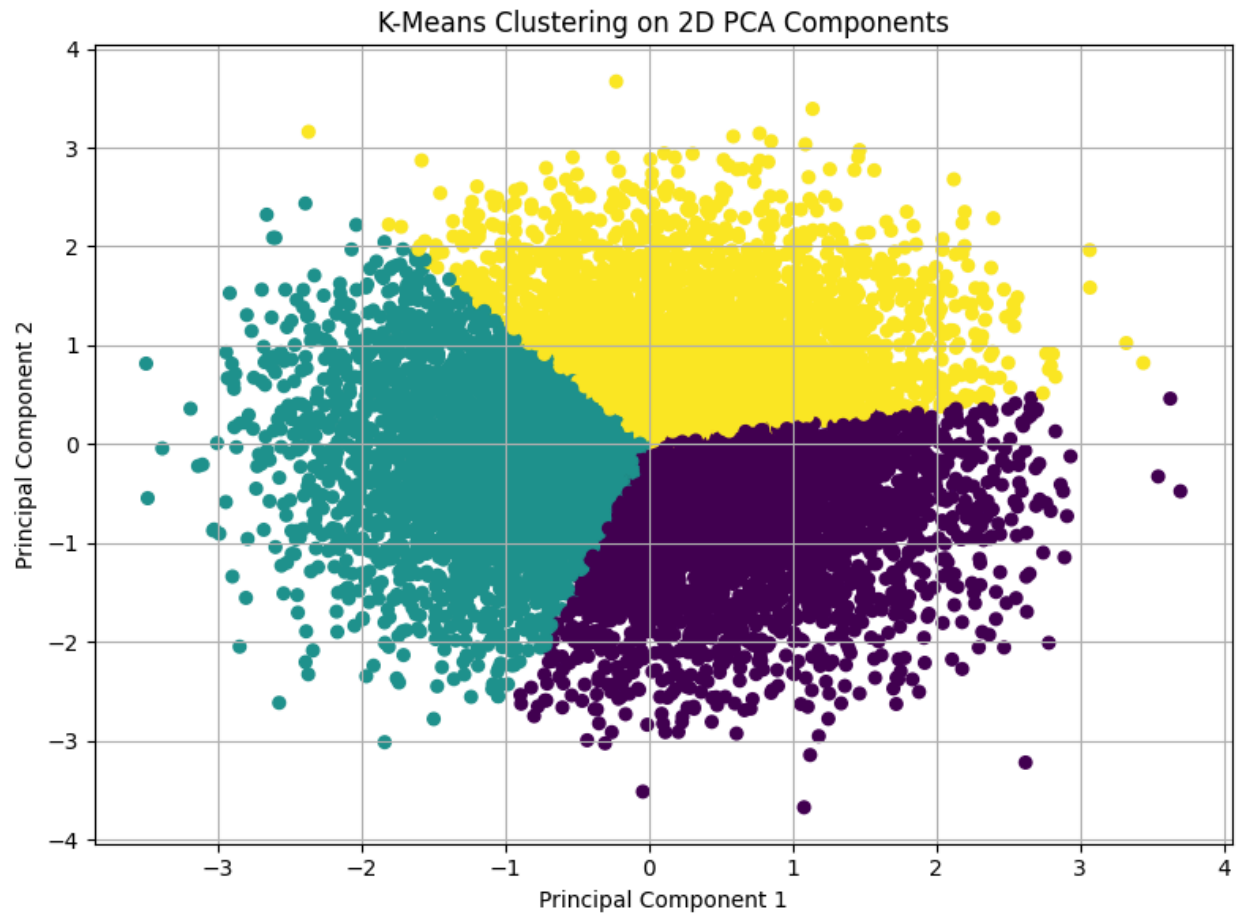


Figure 4.4 K-means Clustering

4.2 Agglomerative Nesting (AGNES)

Agglomerative Nesting is a type of hierarchical clustering method that builds nested clusters by progressively merging pairs of clusters. It starts with each data point as its own individual cluster and then repeatedly combines the closest clusters based on a chosen distance metric, continuing this process until all points are merged into a single cluster or a stopping criterion is met [4]. To find the best number of clusters for Agglomerative Clustering, we used a dendrogram. By drawing a horizontal line around the distance of 30, it cuts through 4 main branches, suggesting that 4 clusters would be a good choice for separating the data clearly. Below is the python code and dendrogram graph.

```
linkage_matrix = linkage(df_pca[['PC1', 'PC2']], method='ward')
plt.figure(figsize=(10, 6))
dendrogram(linkage_matrix)
plt.axhline(y=30, color='r', linestyle='--')
plt.title('Dendrogram for Agglomerative Clustering')
plt.xlabel('Sample Index')
plt.ylabel('Distance')
plt.tight_layout()
plt.savefig('agnes_dendrogram.png')
plt.show()
```

Figure 4.5 Python code for dendrogram

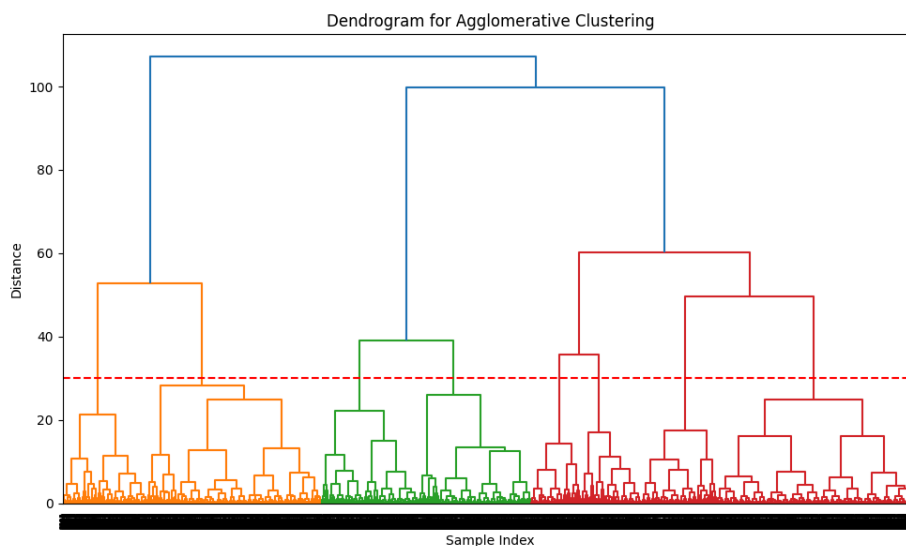


Figure 4.6 Dendrogram for Agglomerative clustering

After the clusters are determined ($k_{\text{agnes}} = 4$), the cluster is used for the AGNES model. Below is the python code for AGNES and the result of the clustering.

```
pca_for_agnes = df_pca[['PC1', 'PC2']].copy()
agnes = AgglomerativeClustering(n_clusters=k_agnes, linkage='ward')
agnes_labels = agnes.fit_predict(pca_for_agnes)
df_pca['agnes_cluster'] = agnes_labels
```

Figure 4.7 Python code for Agglomerative clustering

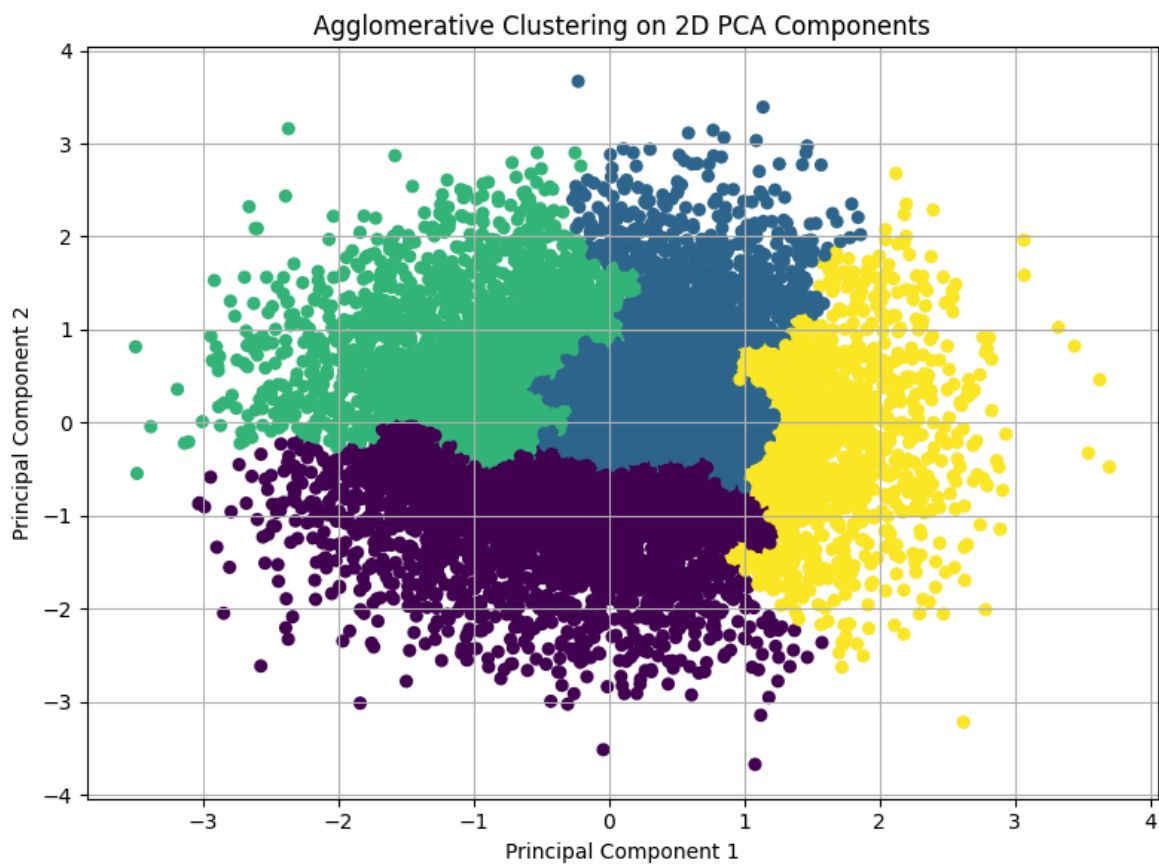


Figure 4.8 Agglomerative clustering

5.0 Discussion

5.1 Evaluation Metrics

The discussion focuses on explaining the evaluation metrics which are Silhouette Score, Calinski-Harabasz and Davies-Bouldin Score. These metrics assess the effectiveness of each model in clustering analysis. These metrics provide a more comprehensive view of the model's strengths and weaknesses. This comparison will provide us with meaningful insight regarding each clustering algorithm capability and similarity between objects. There is also the summarization for each cluster K-Means and AGNES (Agglomerative Nesting) algorithm.

Algorithm/ Result	Silhouette Score	Calinski-Harabasz	Davies-Bouldin Score
K-Means	0.3307	6047.3392	0.9370
AGNES	0.2684	4763.8509	1.0863

Table 5.1: Comparison Between K-Means and AGNES for Silhouette Score, Calinski-Harabasz, Davies-Bouldin Score

Looking at the scores, the K-Means clustering model did a better job of grouping your heart disease data compared to the AGNES model. The numbers show that K-Means created clearer, more separate, and tighter groups, making it the stronger choice for understanding your data's natural segments.

5.2 Clusters Summaries

To explore the detailed insight for the clusters, the profile of each cluster was determined by analyzing the K-Means and AGNES Cluster Centroids, which represent the mean values of all features within that specific cluster.

K-Means Cluster Centroids (in original feature space):									
	Age	Gender	Blood Pressure	Cholesterol Level	Exercise Habits	Smoking	Family Heart Disease	Diabetes	
0	55.862	0.338167	147.297	235.3	0.977541	0.611529	0.468438	0.5	
1	41.654	0.631547	151.056	208.983	1.10603	0.583401	0.699878	0.3	
2	50.5687	0.534787	150.917	232.513	0.950798	0.344129	0.315661	0.5	
Agnes clusters found: [0 1 2 3]									
Agnes (Agglomerative) Cluster Centroids (in original feature space):									
	Age	Gender	Blood Pressure	Cholesterol Level	Exercise Habits	Smoking	Family Heart Disease	Diabetes	
Cluster_0	48.1177	0.464382	148.49	218.247	1.07637	0.695342	0.686817	0.5	
Cluster_1	51.6356	0.48736	149.99	232.448	0.962852	0.42097	0.367801	0.5	
Cluster_2	41.047	0.694174	152.49	211.884	1.06871	0.434816	0.572664	0.3	
Cluster_3	61.8665	0.263973	146.975	250.424	0.880696	0.476912	0.234344	0.5	

Figure 5.1 Result of K-Means and AGNES Cluster Centroids

5.2.1 K-Means Clusters Summaries

1. Cluster Name: Older Women with Many Risks

These are mostly older women (around 56 years old). They are dealing with lots of health issues like high blood pressure, high cholesterol, and high triglycerides, plus a high body mass (BMI). Many of them smoke and drink quite a bit. They are also pretty stressed, have high blood sugar, and their bodies show signs of a lot of inflammation, putting them at high risk for heart problems.

2. Cluster Name: Younger Men Facing Early Risks

This group is made up of younger men (around 42 years old). A big red flag for them is that heart disease runs in their family. Even though they might exercise, they tend to drink a lot of alcohol and consume a lot of sugar. They are often stressed out and do not get enough sleep, leading to high triglycerides and blood pressure. These habits could cause heart issues to show up early.

3. Cluster Name: Stressed Adults with Diabetes

These are middle-aged folks (around 51 years old), a mix of men and women. Their main health battles are diabetes and really high stress levels. They seem to handle smoking and drinking better than some other groups, but keeping their diabetes in check and finding ways to de-stress are crucial for them.

5.2.2 AGNES Clusters Summaries

1. Cluster Name: Complex, High-Risk Women

These are mostly mid-aged to older women (around 48 years old). Heart disease runs strong in their families, and they are very likely to have diagnosed high blood pressure. A lot of them smoke and drink heavily. Their blood also shows high triglycerides and inflammation, meaning they need a lot of support for their health.

2. Cluster Name: Stressed-Out Women

This group is primarily middle-aged women (around 52 years old). What stands out most for them is their extremely high stress levels. Even though they are not heavy smokers or drinkers, they still struggle with high cholesterol and triglycerides. It really highlights how much stress can impact their heart health.

3. Cluster Name: Younger Men with Tricky Habits

These are our youngest folks (around 41 years old), mostly men. They tend to consume a lot of sugar and alcohol, and they are getting the least amount of sleep. Despite exercising, these lifestyle choices are leading to high triglycerides and other early signs of heart risk. They need a wake-up call about their daily habits.

4. Cluster Name: Older Women with Metabolic Issues

This is the oldest group (around 62 years old), mainly women. Their biggest health concerns are very high cholesterol, high BMI, and a high chance of having diabetes. While they might not smoke or drink as much, their age and existing metabolic conditions mean they need strong focus on managing these chronic problems.

5.3 Business Insights Evaluation

5.3.1 Heart Disease Status Segmentation

Based on the evaluation metrics, K-Means performs better than AGNES across all scores, indicating it is more effective in clustering the data. K-Means has a higher Silhouette Score, showing better-defined groupings, and a higher Calinski-Harabasz score, which reflects more compact and well-separated clusters. Additionally, it has a lower Davies-Bouldin score, meaning there is less similarity between clusters. Overall, K-Means demonstrates stronger clustering quality compared to AGNES.

In the context of segmenting heart disease status, these results suggest that K-Means is more suitable for identifying distinct patient groups based on health attributes. The better-defined clusters generated by K-Means can represent different health risk categories such as low, medium, or high risk of heart disease more accurately. This enables healthcare practitioners to better analyze and target interventions or treatments based on the specific characteristics of each group.

5.3.2 Lifestyle-Driven Heart Disease Risk & Preventative Program Design

This analysis shows that different groups of people have unique heart disease risks, mainly because of their lifestyles. This means we can create more effective prevention plans:

- **Older Women:** Those who are older, have a high BMI, metabolic issues, and use tobacco or alcohol need help managing their metabolism and getting support for addiction.
- **Younger Men:** Younger men with poor sleep, high sugar and alcohol intake, and a family history of heart disease would benefit from early changes to their habits and stress reduction techniques.
- **Stressed Middle-Aged Adults:** Adults in middle age who experience high stress, along with diabetes and high triglycerides, can be helped by programs focusing on mental health and managing chronic diseases.

By matching prevention strategies to these specific groups, public health efforts can better tackle controllable risks and reduce the overall impact of heart disease.

6.0 Conclusion

This project has successfully applied both of the clustering algorithms including K-means and AGNES. The final result shows that K-Means is the more effective algorithm compared to AGNES for identifying meaningful groups in the dataset. K-Means achieved better performance across all evaluation metrics with a higher Silhouette Score and Calinski-Harabasz index, and a lower Davies-Bouldin score indicating stronger, more distinct, and well-separated clusters.

The K-Means clusters provide clearer insights, such as identifying older women with multiple health risks, younger men with emerging lifestyle-related issues, and middle-aged adults struggling with stress and diabetes. These groupings are not only statistically valid but also practically useful for understanding heart disease risk profiles. While AGNES also identifies important patterns, its clusters appear less compact and more overlapping, reducing clarity in segmentation.

To make a further improvement, here is the approach that can be reconsidered. The approaches are refining feature selection and engineering, scaling and normalization, hybrid clustering methods and taking validation from domain experts such as healthcare professionals.

Overall, this report concludes that K-Means offers a stronger foundation for health risk segmentation, and with additional refinement, it can become an even more powerful tool for early detection, personalized care, and targeted health interventions in heart disease prevention.

7.0 Reflection

7.1 MOHAMED ALIF FATHI BIN ABDUL LATIF

7.1.1 PART A

For project 1 (classification), my overall contribution is making a conclusion, data preprocessing and sampling. For the classification algorithm, I incharge to making the XGBoost algorithm and part of the K-NN algorithm including making an evaluation matrix for determining accuracy, precision, recall and F1-Score for all the insights. For project 2 (clustering), my overall contribution is making AGNES clustering algorithms. For discussion, my part is making summaries of both clustering algorithm results using cluster centroid and conducting one insight.

One of the challenges of doing these projects is dealing with imbalanced datasets which can lead to initial models biased towards the majority class. Not knowing which one is correct or not when making a data cleaning and applying algorithms is also one of my challenges doing this project as there are so many methods for data cleaning and classification and clustering algorithms.

Now, after completing the projects, I learned how to correctly use machine learning to determine or evaluate some valuable information. I learned how to do classification and clustering algorithms using python.

7.1.2 PART B

This Data mining course provided me with a comprehensive and hands-on introduction to machine learning which is covering both theoretical and practical implementation. This helps me to understand more about machine learning algorithms, data preprocessing and evaluation models because I need to do the task practically, not only by listening to lectures.

Through all assessments in data mining, I analyze all the techniques with practical learning and produce solutions from given data mining problems and business opportunities which I think have met the intended learning outcomes.

New skills that I learned from this course are able to use python for machine learning tasks and more accurately discover the knowledge inside the dataset. I learned many python libraries like matplotlib, scikit-learn and pandas in order to do classification and clustering. I also learned how to discover patterns and hidden knowledge in a dataset effectively as this is a crucial aspect of data mining.

For future academic or professional contexts in data mining, I will use the knowledge from this course to pursue in data engineering by making some individual projects for me to sharpen my current data mining skills. These projects can be my foundation for my future learning as I maybe move to the data science field.

7.2 MUHAMMAD AFIQ DANIAL BIN ROZAIDIE

7.2.1 PART A

Overall tasks that I do in Project 1 (Classification) are making the introduction for the report, involved in cleaning and sampling the dataset and doing Naive Bayes and also taking some part of the KNN classification model for all the business insights. These classification models also include evaluating a confusion matrix and performing a metric score like accuracy, precision, recall and F1-Score. While for Project 2 (Clustering), the contribution that I do is to also write an introduction, involved in making data preprocessing and preparation like handling missing data, mapping, normalization and reducing dimensionality. The algorithm that I conduct is the K-Means algorithm and do the evaluation metrics. Lastly, I provide one of the insights and make a conclusion in project 2 report writing.

One of the challenges I faced in both projects was dealing with the uncertainty of whether the dataset, after the cleaning process, was of good quality or still contained hidden issues. Not to mention that the dataset we got also contains some of the imbalanced data or bias.

After finishing both of the projects, I learn on how to clean the dataset like fill in the missing data, standardize the dataset and evaluate a performance for each modelling whether using Python or Rapid Miner.

7.2.2 PART B

My sharing on the Data Mining course is that I have gained valuable insights into the end-to-end process of extracting meaningful patterns and knowledge from large datasets. The hands-on both Project 1 and Project 2 helped me understand how to handle real-world data challenges, such as missing values, unbalanced classes, and feature selection.

In this Data Mining course, there are three learning outcomes which include analyzing the processes and techniques used in data mining, producing data mining solutions for a given problem and proposing business opportunities and solutions in data mining. I can say that I have successfully achieved the intended learning outcome.

The new skill that I learned is how to apply essential data preprocessing techniques, explore data visually, and implement both supervised and unsupervised machine learning algorithms such as classification, clustering, and association rule mining. All of these were accomplished using Python as the primary programming language throughout the project.

In future academic projects, I will use data preprocessing, clustering, and classification techniques to analyze complex datasets more effectively and draw meaningful conclusions. Professionally, I aim to use data mining tools to support data-driven decisions, whether in marketing, healthcare, or financial analysis.

8.0 References

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